

Mock Viva — likely questions, strongest answers

Ordered by how likely each is to be asked. The bold line under each answer is the sentence to land on if you only get one; the red line is the way that answer can go wrong. Say these out loud before the day — the gap between having the answer and delivering it is wider than it looks on paper.

Tier 1 · almost certain

"Summarise your project in two minutes."

The question is whether models trained separately learn the same thing. I narrowed it to something testable: are their representations related by a single linear map? I built two systems to find out. The first connects two vision encoders with one 0.79 MB matrix and reaches 95 percent of a jointly-trained ceiling — and when I replaced that matrix with a neural network, it converged to exactly the linear answer, twice. So the relationship really is linear, not just approximable by something linear. The second is a shared 512-dimensional space that seven encoders enter, where a component trained on one works on encoders it never saw at 93 to 96 percent. The scientific finding underneath is that what they share is local: they agree strongly about which items are neighbours and much less about the global shape of the space.

Land on: "They don't share coordinates — raw agreement is exactly chance — but they describe the same world, and the translation is one matrix small enough to fit on a phone."

Trap: Don't recite nine findings. Two systems, one scientific result, stop.

"Isn't whitening cheating? You fixed the data until it worked."

Whitening uses no labels and no pairs — it is fitted on training targets only and it is linear. So it is an instance of the thesis, not an exception to it: the full map is whitening composed with ridge, which is still one matrix. And it was not a fix applied until something worked — it was a prediction. Section C.11 said a collapsed space should be rescuable by a coordinate change, and C.14 then predicted the ordering of two encoders I had not yet tested. That prediction was written down before the run and came back at rank correlation +1.00.

Land on: "It's label-free, it's linear, and it made a prediction before it was tested."

Trap: If you sound defensive, it reads as a patch. It's a mechanism — say it as one.

"How do I know your interpretations aren't motivated reasoning?"

Because seven of them turned out to be wrong and are still in the report. That collapse would depress shape correlation — false, rank measures survive it. That the hub discards a collapsed space's directions — false, it reconstructs them perfectly. That reach into the hub predicts writability — false, effective rank does. That adding encoders improved transfer — overstated, the gain was six queries in a thousand. That the ratio-to-chance collapse proved locality — partly a normalisation artifact, corrected. And both explanations of the hub width cliff, including my own preferred one, failed the same test.

Land on: "A claim that survived a test which could have killed it is worth more than one never exposed to one."

Trap: Don't list all seven. Give two, then offer the rest if they want them.

Tier 2 · very likely

"What would have falsified your thesis?"

Several things, and each was a real risk when I ran it. If the MLP had beaten the linear map, 'related by a linear transformation' would have been wrong — it tied to four decimals instead, twice. If the random-map control had scored above chance, the transfer result would have been an artifact of the head rather than the hub — it collapsed to chance. If cross-modal retrieval had sat at chance, convergence would not cross modality — it reached 41 to 58 percent of a matched ceiling. And the pre-registered gate at 90 percent of ceiling was not met; it is reported as not met, at 89.0.

Land on: "The gate that failed is in the report at 89.0 percent. A gate that never fires is decoration."

Trap: Naming a falsifier you never actually risked sounds rehearsed. Use the ones that could have gone the other way.

"Your models are all transformers. Isn't this just an architecture artifact?"

It was the strongest open objection; it is now closed on two fronts. Lineage: SigLIP 2 shares no DINOv2 ancestry and joins the frozen hub at 94.2 percent, inside the 93.8-96.5 sibling band. Architecture: ConvNeXt-base - a convnet, trained on 22,000 labels, differing on architecture AND supervision - transfers at 96.7 percent, ABOVE the transformer band (E.2). E.12 sharpens it: ConvNeXt's raw geometry agrees LEAST with every encoder, yet transfers best. Caveat I raise myself: ConvNeXt borrows transformer-era design; a ResNet would be a purer control.

Land on: "A convnet with no shared ancestry and the weakest raw agreement transfers above the transformer family. Architecture does not bound the claim - E.2, E.12."

Trap: Never say 'I have not closed it' - you have. Volunteer the ResNet caveat before it's used against you.

"Verification AUC 0.999 — that looks like overfitting."

It is held out — fitted on 3,000 pairs, scored on 1,000 images the fit never saw. But the decisive evidence is internal: the same map scores 0.999 against a random impostor, 0.848 against the hardest of a thousand, and R@1 0.987 when one of a thousand must be picked. Overfitting inflates everything. A memorised map does not drop 0.15 when the negative gets harder. The 0.999 is high because that question is easy — a random impostor is usually a dog against a kitchen.

Land on: "Three difficulties, one system. Difficulty of the negative explains the spread, not memorisation."

Trap: Never quote 0.999 alone. Volunteer the other two before you're asked.

"Why does your report say the hub 'works' when agreement is only local?"

Because those are two different components of the same structure, and I measured which one the hub uses. Across a width sweep, cross-encoder local agreement falls monotonically from 0.603 to 0.462 while transfer rises from 0.350 to 0.489. They move in opposite directions — so transfer cannot be running on local structure. It runs on the global linear component, which is real but thin. That explains both facts at once: the hub works because the global part exists, and it never reaches the ceiling because the rich local part is not what a linear map transports.

Land on: "A thin global signal and a persistent retrieval shortfall are the same fact seen twice."

"Your text encoders top out at bge/BERT/GPT-2 scale (768-1024-d) - Huh et al. go to 70B-parameter LLMs. Isn't that a different regime?" Yes, and I'd rather concede it than have it drawn out of me. The effect is already clearly measurable at this scale, and the closed-form, CPU-only design is the point of this project, not an accident of scale - every result here runs without a GPU past the one-time encoding pass. Scaling the text side is a clean future-work item, and PRH's own capacity-ladder plot predicts it should only strengthen the result. One thing I do not claim: that alignment predicts downstream capability - that correlation is Huh et al.'s finding, not mine. My evidence is about the geometry, not the benchmark.

Land on: "Huh et al. showed alignment rises with scale and correlates with capability; I took the hypothesis as given and asked what is actually shared, whether one matrix can carry it, and where it fails - and the answers are local structure, yes at 96.7 percent, and at the source encoder's rank."

Trap: Don't claim your data shows alignment predicts capability - you didn't measure downstream task performance. Cite Huh for that link.

Tier 3 · likely if they read closely

"The width cliff - transfer collapses between 512 and 768 dimensions. What is it?"

A numerical artifact of the fit, resolved to one direction - after seven accounts were falsified. The head's coordinates have algebraic rank 768; at hub width 768 the last singular value is near-null, 0.0054 against 10.79 a step earlier, and least squares divides by it. Transfer holds at 0.894 to $k=760$, then 0.009 at 768 - a step, not a decline. Pre-registered, confirmed out of sample on DINOv2-base; truncation or a larger head alpha repairs it. Two falsified accounts were mine, including a rank test that compared entropy rank to a prediction about algebraic rank - recorded as an error.

Land on: "Seven falsified, the eighth pre-registered and confirmed to one singular value. A division to avoid, not a property of shared representations. Rule: hub width below the source's rank."

Trap: Never say 'unexplained' or 'open'. Volunteer G13 - your own falsified test is the strongest thing on this page.

"Recent work says similarity metrics are confounded by model width. Does that affect you?"

I tested it directly. Using a permutation null — shuffling one side's rows so widths and geometry are preserved but the correspondence is destroyed — my neighbourhood metric's null lands exactly on k/N , so the primary result needed no correction. CKA does have a width-dependent null, about +0.094, rising with embedding width. Correcting it actually STRENGTHENS my secondary finding: the widest pair, the two DINOv2 siblings, was inflated most, so the gap between the shared-objective pair and the shared-lineage pair widens from +0.024 to +0.181.

Land on: "The confound was biasing the comparison in favour of the lineage pair. Correcting it made the objective result stronger."

"Why bge-m3 as the target if SBERT scores better?"

Comparability. Every ceiling and every percentage-of-ceiling figure in the project is measured against bge. Switching in the final sections would make those numbers incomparable for a gain of about three points. SBERT is genuinely the better target — 101.0 percent writable against 98.1, raw $R@1$ 0.510 against 0.466 — and it is offered as an alternative in the demo. It is retained for comparability, not because the data chose it, and the report says exactly that.

Land on: "Kept for comparability, not merit — and stated as such."

"What is the single weakest part of this work?"

The causal claim that shared training objective predicts agreement better than shared lineage. Every metric I have orders it the same way, and the calibration made it stronger — but the groups are as small as one pair, and the pairs are not independent since three encoders share a teacher. I report it as exploratory description and explicitly not as a test. Its value is that it identifies the next experiment: more encoders spanning objectives, not more spanning sizes.

Land on: "It's the most interesting hint in the project and the least powered. Both of those are in the report."

Trap: Naming a trivial weakness reads as evasion. Name the real one — it buys credibility for everything else.

"How well-conditioned is the hub basis itself? How many rows estimated it?" The honest number is tight, and I would rather give it than be asked. The hub PCA is estimated on 8,533 training rows over a 5,376-dimensional concatenation - aspect ratio $\gamma = d/n = 0.63$, and 0.90 for the seven-space version at 7,680 dims. The covariance stays full rank, since n is still greater than d , but near $\gamma = 1$ the sample spectrum spreads away from the true one. Two things cut the other way. Only the top 512 directions are kept, about seven percent of the spectrum, and leading eigenvectors are estimated far more reliably than the bulk. And whatever distortion remains biases transfer DOWNWARD, so the published figures are conservative. The fix is costed: a 35,133-item corpus brings γ to 0.22. It is in the Atlas as future work.

Land on: "Gamma is 0.63 for the published hub and 0.90 for the seven-space one - high, but full rank, and only the best-estimated 512 directions are kept. The bias runs conservative, and re-estimating on 35k items is costed and queued."

Trap: Do not quote the ≥ 5 rows/dim floor here - that is the REGRESSION rule and this is a PCA. Using it invites a correction you should have made yourself. Give gamma.

Tier 4 · possible

"Why closed-form ridge instead of training a mapper?"

Because the claim is about the relationship, not about what a sufficiently powerful network can learn. A trained mapper could succeed by learning the task itself, which would prove nothing about convergence. Closed form removes that ambiguity entirely — there are no gradients, no epochs, no capacity to hide behind. And the MLP contrast is the deliberate exception that shows what capacity buys: exactly nothing, to four decimals.

Land on: "Closed form means the result cannot be explained by the mapper's own learning."

"What is effective rank and why do you use it everywhere?"

It is the spectral entropy of a space's covariance — roughly, how many directions actually carry variance, as opposed to how many the space nominally has. GPT-2's caption space has 6.6 effective directions out of 768. It matters because it predicts almost everything else in the project: which spaces are writable, where cosine becomes meaningless, and which pairs the hub rescues. The encoder ladder was ordered by it before the run and came back correct.

Land on: "It is the diagnostic that predicted, not just described."

"Could the whole thing be explained by shared training data?"

Partly, and I bound it rather than dismiss it. The image and text encoders never trained on each other's modality, so cross-modal agreement cannot come from shared examples. But all seven were trained on broadly web-scale corpora, so some overlap in content is certain. What I can say is that agreement tracks training OBJECTIVE more closely than lineage or modality, which is not what a pure shared-data account predicts.

Land on: "It cannot be ruled out, and the objective result points away from it."

"Is any of this useful outside a coursework project?"

The engineering claim is that many small expert models joined by a shared hub can behave as one system: N maps instead of N squared, adding a model is one closed-form solve, and a component trained on one encoder serves the rest at 93 to 96 percent. Whether that beats one large general model at matched parameters is the benchmark I would run next, and I have specified it with a pre-registered gate.

Land on: "The components exist; the benchmark is specified and not yet run."

Three habits for the room

Volunteer the caveat before they find it. Every strong answer above ends by naming its own boundary. That converts an objection into evidence of care.

Quote against a control, always. No number in this project stands alone — 0.999 against chance 0.481, 39.7 percent against 595 times chance, 94.2 percent against a 93.8 to 96.5 band. Saying the control aloud is what separates a measurement from a claim.

If you don't know, say the shape of the answer. "I haven't measured that. What I would do is X, and here is what would count as a yes." That is a stronger response than a guess, and this project has earned it — seven times over.